A378581

Square-index residues for theta product substitution

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

Let $\Theta = 1+2\sum_{j\geq 1} x^{j^2}$. The unique constant-one integral solution of $A(xA) = \Theta$ satisfies $A \equiv \Theta \pmod 4$. A division-free construction and a square-zero perturbation argument prove the full positive-index classification.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A378581 [2], retrieved on 24 October 2026. Its initial sequence index is 0. The constant coefficient is one; the residue classification below is restricted to positive indices.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1\rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

In this manuscript $\Theta(x) = 1 + 2\sum_{j\geq 1} x^{j^2}$. The chosen source is $A(xA) = \Theta$. The auxiliary construction treats both signs.

2 Formal algebra used in the proof

Lemma 2.1 (Coefficientwise construction)**.** *Let K be a commutative ring. Suppose an operator Φ preserves normalized series and, for $r \geq 2$, satisfies*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}).$$

Then Φ has exactly one normalized fixed point in $K[[x]]$. More generally, a residual equation whose degree-r condition determines the next unknown coefficient with multiplier one has a unique solution once its initial coefficients are fixed and its lower-degree conditions hold identically.

Proof. Begin with $G_0 = x$ and set $G_{m+1} = \Phi(G_m)$. Normalization gives $G_1 - G_0 = O(x^2)$. Inductively, $G_{m+1} - G_m = O(x^{m+2})$, so each coefficient eventually stabilizes. Define G using those stable coefficients. The displayed implication ensures that each coefficient of $\Phi(G)$ equals

the corresponding stable coefficient, hence $G = \Phi(G)$. If two fixed points agree below degree r, they agree through degree r; induction proves uniqueness.

For the residual version, set the next coefficient to the negative of the known residual term. Its multiplier is one, so the construction uses only ring operations. No earlier equation changes, and every equation is eventually imposed. The same coefficient comparison proves uniqueness. □

Lemma 2.2 (Units and reversion)**.** *Every constant-one series has a unique constant-one multiplicative inverse. Every normalized series has a unique normalized two-sided compositional inverse. These inversions, formal differentiation, and valid substitution commute with coefficient ring homomorphisms.*

Proof. For $UV = 1$ with $U(0) = 1$, the coefficient of degree $n > 0$ is v_n plus a finite expression in earlier coefficients of V, which determines v_n without division. For normalized G, the coefficient of degree $n > 1$ in $G(H)$ is h_n plus an expression in earlier coefficients of H, giving a unique normalized right inverse H.

To see the second inverse identity, construct a normalized right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, and hence $H \circ G = x$. The same association proves uniqueness. Each coefficient of a valid composition is a finite polynomial in coefficients. Differentiation multiplies the degree-$(n+1)$ coefficient by $n+1$. Ring homomorphisms therefore preserve these operations and equations; uniqueness identifies the mapped inverses with the inverses of the mapped series. □

Lemma 2.3 (Doubling and substitution)**.** *For $U, V \in (\mathbb{Z}/4\mathbb{Z})[[x]]$,*

$$2U = 2V \quad \Longleftrightarrow \quad U \equiv V \pmod 2.$$

If B, D have zero constant coefficient and $B \equiv D \pmod 2$, then $2H(B) = 2H(D)$ for every series H over $\mathbb{Z}/4\mathbb{Z}$.

Proof. For residues u, v modulo four, $2u = 2v$ exactly when their reductions modulo two agree. Apply this fact to each coefficient. Reduction modulo two commutes with substitution, giving the second assertion. In particular, this lemma never cancels two inside $\mathbb{Z}/4\mathbb{Z}$. □

3 Construction and square support for A378581

3.1 A general integral construction

To treat the selected source, we construct a pair of auxiliary substitution equations. For a constant-one target, these equations are

$$A_-(x/A_-(x)) = \Theta(x), \qquad A_+(xA_+(x)) = \Theta(x), \qquad A_\pm(0) = 1. \tag{1}$$

We first solve them with an arbitrary constant-one target T in place of Θ, over any commutative coefficient ring K.

Lemma 3.1. *For each $T \in K[[x]]$ with $T(0) = 1$, there is exactly one normalized G satisfying*

$$G \circ G = GT. \tag{2}$$

Proof. Let $E(G) = G \circ G - GT$. Its coefficients in degrees zero and one vanish for every normalized G. Suppose $G - H = dx^r + O(x^{r+1})$, with $r \geq 2$. Decompose the difference of self-compositions as

$$G \circ G - H \circ H = (G - H) \circ G + (H \circ G - H \circ H).$$

Because $G = x + O(x^2)$, the first summand is $dx^r + O(x^{r+1})$. Write $H = x + x^2W$. The second summand is $(G - H) + G^2W(G) - H^2W(H)$. The difference after $G - H$ has order at least $r + 1$: factor $G^2 - H^2$, and use that substitution in any series preserves an agreement modulo x^r. Thus the second summand also is $dx^r + O(x^{r+1})$.

Finally, $(G - H)T = dx^r + O(x^{r+1})$, since $T(0) = 1$. Therefore

$$E(G) - E(H) = dx^r + O(x^{r+1}).$$

The degree-r residual has multiplier $2 - 1 = 1$ on the next unknown coefficient. Lemma 2.1's triangular construction proves existence and uniqueness with no division, over the stated ring. □

Write this unique G as $G = xU$. Then $U(0) = 1$ and

$$G \circ G = G\,U(G) = GT.$$

Cancelling G is valid even when K has zero divisors: multiplication by x is injective coefficient-twise, and U is a unit. It follows that $U(xU) = T$. Conversely, any constant-one solution of $U(xU) = T$ gives a normalized solution xU of (2). Consequently there exists exactly one such U; denote it by $\mathrm{Plus}(T)$.

Define

$$\mathrm{Minus}(T) = \mathrm{Plus}(T^{-1})^{-1},$$

where both occurrences of inverse here are multiplicative unit inverses. If $U = \mathrm{Plus}(T^{-1})$, then $U(xU) = T^{-1}$. Substitution respects products, so $U^{-1}(xU) = T$. Since $x/\,\mathrm{Minus}(T) = xU$, this is

$$\mathrm{Minus}(T)(x/\,\mathrm{Minus}(T)) = T.$$

Conversely, if constant-one V satisfies $V(x/V) = T$, then inversion of that identity gives $V^{-1}(xV^{-1}) = T^{-1}$. The uniqueness of $\mathrm{Plus}(T^{-1})$ identifies $V^{-1} = U$ and hence $V = \mathrm{Minus}(T)$. We have proved existence and uniqueness for both source equations over every commutative ring, including $\mathbb{Z}$.

The quotient is literal: $V(x)\,(xV(x)^{-1}) = x$. It is not interpreted as a compositional inverse. Coefficient maps preserve the two source equations and their unit inverses, so uniqueness also proves that Plus and Minus commute with reduction of coefficients.

3.2 The common square-support congruence

Theorem 3.2. *Let $A_- = \mathrm{Minus}(\Theta)$ and $A_+ = \mathrm{Plus}(\Theta)$ over $\mathbb{Z}$. These are the unique constant-one integral solutions of* (1)*, and*

$$A_-(x) \equiv A_+(x) \equiv \Theta(x) \pmod 4.$$

In particular, for either sign and every $n > 0$,

$$[x^n]A_\pm \equiv \begin{cases} 2 \pmod 4, & n = j^2 \text{ for some } j \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Work over $\mathbb{Z}/4\mathbb{Z}$ and put $S = \sum_{j \geq 1} x^{j^2}$, so $\Theta = 1 + 2S$. The series $B = x\Theta$ has zero constant coefficient and satisfies $B \equiv x \pmod 2$. Lemma 2.3 gives $2S(B) = 2S(x)$. Therefore

$$\Theta(x\Theta(x)) = 1 + 2S(x\Theta(x)) = 1 + 2S(x) = \Theta(x).$$

The uniqueness just proved identifies $\mathrm{Plus}(\Theta) = \Theta$ over this ring. Also

$$\Theta^2 = (1 + 2S)^2 = 1 + 4S + 4S^2 = 1,$$

so $\Theta^{-1} = \Theta$. It follows that

$$\mathrm{Minus}(\Theta) = \mathrm{Plus}(\Theta^{-1})^{-1} = \mathrm{Plus}(\Theta)^{-1} = \Theta^{-1} = \Theta.$$

Compatibility with coefficient reduction identifies these series with the reductions of the canonical integer solutions. Finally, the coefficient of Θ at a positive index is two exactly when the index is a square, and zero otherwise. □

This proves the coefficient conjecture for the selected source, with the other substitution sign supplied by the same general construction [1, 2]. Each equivalence is accompanied by the complementary integer statement $4 \mid a_n$ at nonsquares. The constant term is one in both sequences; excluding $n = 0$ is essential. The modular argument only uses the form $1 + 2S$ and $S(0) = 0$, so its mechanism is more general than square support. The shared mechanism is included to make this manuscript self-contained.

4 Separate Lean proof and checked scope

The entry file is `lean/Entries/A378581.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A378581`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `theta_constant_term`.
- `theta_coefficient_formula`.
- `plus_constant`.
- `plus_source`.
- `plus_exists_unique`.
- `plus_unique`.
- `plus_series_mod_four`.
- `plus_congruence`.
- `plus_square_iff`.
- `plus_four_dvd_iff`.

The main implementation module is `Round3.ThetaSubstitution`. The project pins Lean 4.19.0 and Mathlib `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The supplied collection report records fresh source compilation and transitive axiom audits. All coefficient conclusions are unbounded.

Limits and status. The multiplicative unit inverse and compositional inverse are distinct. Analytic convergence and real special values are not proved. No global novelty, exclusive priority, peer review, or OEIS acceptance is asserted.

References

[1] Paul D. Hanna, *OEIS A378580*, 8 January 2025. `https://oeis.org/A378580`. Retrieved 24 October 2026.

[2] Paul D. Hanna, *OEIS A378581*, 8 January 2025. `https://oeis.org/A378581`. Retrieved 24 October 2026.